The London Gold Fix Drift

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October 2026

Abstract

Gold futures fall into the LBMA AM gold auction: a short from 10:25 to 10:31 London time earned +1.6 bp per day on average over 4,061 trading days of CME gold futures (2010–2026, $t \approx 17$, 59% of days positive). The effect disappears on English bank holidays, when no auction is held (−0.07 bp, $t = -0.13$, 88 days), which ties it to the auction rather than to the time of day. It is strongly regime-dependent: near zero in the quiet years 2017–2019, weak in 2022–2024 and at its strongest in 2026.

Implemented on a retail gold CFD with a $4 stop, a post-fix rebound leg and a re-entry after stops, the strategy earned +149 R over 489 trades from July 2025 to October 2026 (+0.31 R per trade, about 9 R per month), with a profit factor of 2.55 and a maximum drawdown of 4.3 R. Here 1 R is the loss at the stop, about $250 per trade, or 0.3–0.7 lots. It has traded live since September 2026; its first week lost 0.69 R, in line with the backtest's 0.61 R over the same days. The main risks are the effect's dependence on volatility and the small number of truly independent observations behind any refinement.

AI disclosure. The author used Claude (Anthropic), an AI assistant, to write the data-analysis code, to draft and edit the text, to assist with wording and structure, and as a consultative tool for discussing methodological and technical questions. The research questions and the decisions on what to test and what to accept are the author's. The author reviewed, verified, and edited all AI-assisted content and takes full responsibility for the content of the paper, including its analysis, results, and conclusions.

1 Hypothesis and mechanism

The hypothesis is that order flow which must transact at the LBMA AM gold price pushes the market down around the start of the 10:30 London auction, and that the move partly reverses once the auction clears.

The LBMA Gold Price is set twice a day, at 10:30 and 15:00 London time, in an electronic auction. Producers, central banks, ETFs and bullion banks use it as the reference price for physical and contractual settlements. Participants who know they will trade at the fix, and dealers who will fill client orders at it, can pre-hedge in the futures and spot market in the minutes before and during it. If that flow is persistently one-sided, the price should move in its direction as the auction opens and recover afterwards.

The study does not observe who trades, so the mechanism stays a hypothesis. It does make three testable predictions:

1. The drift is concentrated in a short window around the start of the auction.
2. Part of the move reverses after the auction.
3. On days with no auction, the drift is absent.

All three hold in the data (Sections 4 and 5).

The closest academic evidence concerns the old telephone fixing. Caminschi and Heaney (2014) find that GC futures and the GLD fund move in the direction of the PM fixing within the first four minutes after it starts, well before the result is published: information leaks out of the fixing itself. The effect studied here differs in what is known in advance: its direction. On average the price falls, about 70% of the fall comes in the two minutes around the start, 10:29 to 10:31 (Exhibit 3), and the effect holds both under the old fixing (to March 2015) and under the electronic LBMA auction that replaced it.

One obvious source of fix-linked flow points the wrong way. The largest gold ETF, SPDR Gold Shares (GLD), values its gold at the LBMA Gold Price PM, not AM (SPDR Gold Trust, 2026). If ETF hedging drove the drift, the PM auction should show the stronger effect. It does not. Measured the same way, as the raw six-minute short into the auction on GC over 2010–2026 (14:55→15:01 London), the PM move is +1.41 bp per day with $t = 7.3$, against +1.63 bp and $t = 16.8$ at the AM auction: no larger on average and much noisier, since the PM window overlaps the 10:00 New York data releases. This points to other fix-linked flow, such as producer, central-bank or dealer orders.

2 Data

Two independent data sets are used: exchange futures for the long history, and the broker's own prices for the tradable result.

Exhibit 1. Data sets

Data set	Period	Resolution	Used for
CME gold futures (GC), front contract, Databento	Jun 2010 – Sep 2026	1 minute; 1 second and per-contract minutes for checks	the effect, its history and robustness tests
Retail XAUUSD CFD (MetaTrader 5 broker)	Jul 2025 – Oct 2026	1-minute bars and real ticks	the trading backtest and live reconciliation
MT5 economic calendar	2010 – 2026	event times	news and holiday conditioning

All timestamps are converted to London local time, because the auction follows the London clock through daylight-saving changes. The broker's server clock is New York time plus seven hours and switches on US daylight-saving dates, not European ones. For about three weeks a year the two schedules disagree by an hour. An early version of the pipeline used European dates and shifted the fix by one hour in those weeks; this was found by comparing broker and CME prices minute by minute and corrected before the results below.

English and Welsh bank holidays are generated from the gov.uk rules, including one-off holidays (2011 royal wedding, 2012 and 2022 jubilees, 2022 state funeral, 2023 coronation) and moved dates.

3 Method

The core trade is a short at 10:25:00 London (the close of the 10:24 bar) closed at 10:31:00 (the close of the 10:30 bar). Returns are measured in basis points of the entry price so that 2011 and 2026 are comparable despite gold rising from about $1,200 to over $4,000.

For the futures history the trade is studied in two forms: the raw six-minute move, and the move with a stop at 0.1% of price (about $4 at $4,000), which caps the loss of a day at −10 bp. For

the CFD backtest the full strategy is simulated in an event-driven engine on 1-minute bars with tick-level resolution of stops and targets.

Exhibit 2. Strategy legs on the CFD (London time)

Leg	Entry	Exit	Stop / target
Main short	10:25	10:31	stop \$4, target \$6
Rebound long	10:31, only if price fell $\geq 0.05\%$ into the fix	10:46	stop 0.15% of price
Re-entry short	immediately after a main-leg stop	10:46	stop \$6

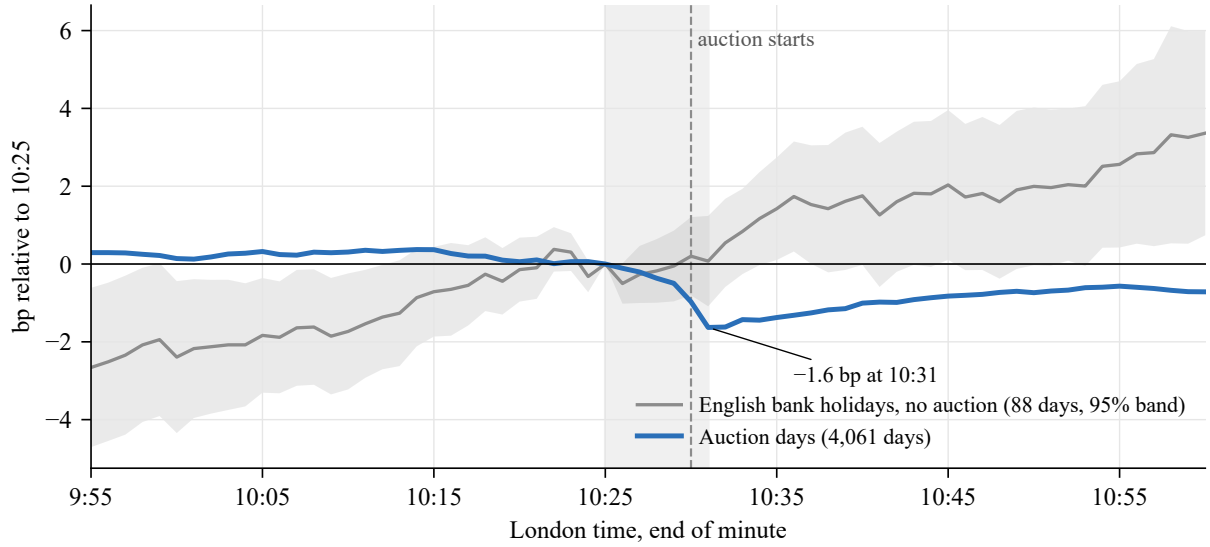
Costs are a fixed \$0.12 spread and \$6 commission per lot. Shorts are stopped on the ask (bid plus spread), not on the bid. A regime filter skips the fix when the 20-day median of morning volatility (sum of absolute minute moves 09:25–10:25) is below 70 bp.

The main statistic is the t-statistic of the daily mean. Every effect is also checked year by year: a real effect should keep its sign in most years, not rest on one or two. Because many hypotheses were tested on the same 16 years, a t-statistic below about 3 is treated as inconclusive, following the multiple-testing hurdle proposed by Harvey et al. (2016). Sharpe ratios are annualised from daily results with a factor of $\sqrt{252}$.

4 Results

From June 2010 to September 2026 the short into the fix earned +1.63 bp per day before costs ($t = 16.8$, 59% of days positive), and +1.53 bp with a 0.1% stop. It is positive in 15 of the 17 calendar years covered (2010 and 2026 are partial), but its size moves with the volatility regime. Before costs the daily series has an annualised Sharpe ratio of 4.2 (4.0 with the stop); by calendar year it ranges from -0.6 to 9.0, with a median of 4.7.

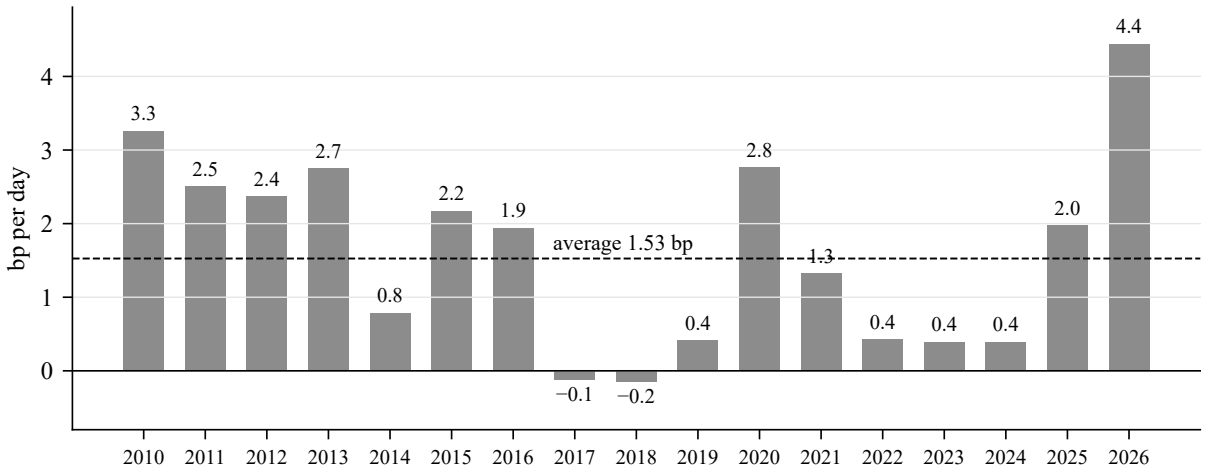
Exhibit 3. Average gold futures price around the AM auction, basis points relative to 10:25 London



Notes:
– Mean of the price at the end of each minute, relative to the close of the 10:24 bar.
– 4,061 auction days and 88 English bank holidays.
– Shaded band: 95% confidence interval for the holiday mean.
– Grey area: the trade window 10:25–10:31.
Source: CME GC front contract, Databento 1-minute bars, June 2010 – September 2026.

On auction days the price is flat until 10:28, then falls by 1.6 bp by 10:31. About 70% of the fall comes in the two minutes from 10:29 to 10:31, around the auction start, and 40% in the auction's first minute alone. About half of the fall is recovered by 10:46 and two thirds by 10:55. On bank holidays the same minutes show no fall: the holiday path drifts up across the morning, but its band contains zero at 10:31.

Exhibit 4. Short 10:25→10:31 London by year, basis points per day before costs



Notes:
– Short with a 0.1% stop; English bank holidays excluded.
– Dashed line: full-sample average.
Source: CME GC front contract, Databento 1-minute bars, June 2010 – September 2026.

The weak periods are 2017–2019, when gold was unusually calm, and 2022–2024. The strongest year is 2026 so far. At $4,000 gold, 1 bp is $0.40 per ounce, so the 16-year average is about $0.60 per ounce per day before costs, against roughly $0.20 of spread and commission on the CFD.

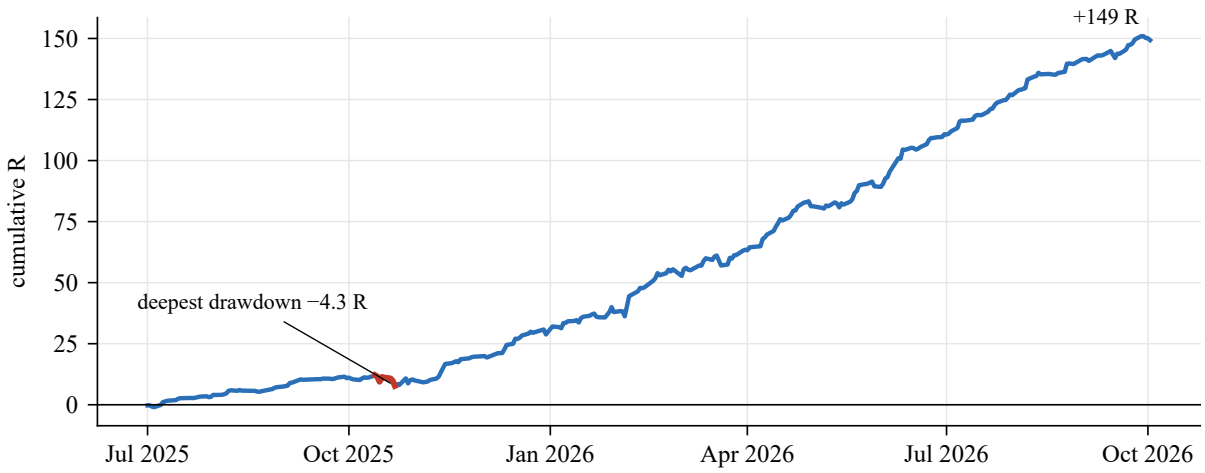
On the CFD, the full three-leg strategy produced the following over July 2025 – October 2026, with every trade sized to risk the same amount, 1 R (about $250, or 2.5% of a $10,000 account).

Exhibit 5. CFD strategy results by leg, July 2025 – October 2026

Leg	Lots (median)	Trades	Net, R	R per trade	Win rate	PF
Main short 10:25→10:31	0.51–0.67 (0.63)	315	+104.7	+0.33	69%	2.81
Rebound long 10:31→10:46	0.30–0.50 (0.38)	132	+28.0	+0.21	60%	2.08
Re-entry short after a stop	0.40–0.43 (0.42)	42	+16.6	+0.40	64%	2.30
Total	–	489	+149.3	+0.31	66%	2.55

Notes:
– Event-driven backtest on broker 1-minute bars; stops and targets resolved on ticks.
– Costs included: $0.12 spread and $6 commission per lot.
– Every trade risks 1 R ≈ $250 at its stop: 5% of a fixed $5,000 sizing base, 2.5% of the $10,000 account.
– Lots vary with the stop distance: $4 on the main leg, $6 on the re-entry, 0.15% of price on the rebound.
– PF: profit factor, gross profit divided by gross loss.
Source: MetaTrader 5 broker XAUUSD data.

Exhibit 6. Cumulative net result of the CFD strategy, in R



Notes:
- Daily net result of all three legs after spread and commission, in R (1 R ≈ $250 risked per trade), summed over time.
- Risk per trade is constant, so the curve shows the edge without compounding.
- Red segment: the deepest peak-to-trough drawdown.

Source: MetaTrader 5 broker XAUUSD data.

The account was up on 69% of trading days and in 14 of 15 full months. The maximum drawdown was 4.3 R, and the worst single day lost 2.05 R. Over all 329 weekdays, counting days without a trade as zero, the daily series has an annualised Sharpe ratio of 7.0 after costs. This period sits inside the strongest regime of the 16-year history, so these figures overstate what a calm year would deliver.

5 Robustness

The strongest evidence that the effect comes from the auction is a natural placebo: on English bank holidays there is no AM auction, and the drift vanishes. Other checks show which refinements are real and which are noise.

Exhibit 7. Fix short on calendar and event days, GC 2010–2026

Check	Days	Short, bp	Others, bp	t, diff.	Verdict
English bank holidays (no auction)	88	−0.07	+1.53	−2.8	effect absent: placebo confirms the mechanism
Mondays	761	+2.16	+1.38	+3.1	stronger; ahead of other days every year since 2019
Wednesdays	830	+1.31	+1.58	−1.2	noise
Fridays	815	+1.27	+1.59	−1.4	noise
1st day of the month	122	+0.66	+1.55	−1.8	inconclusive
30th–31st of the month	197	+1.41	+1.53	−0.2	no effect
Last trading day of the month	196	+1.61	+1.52	+0.2	no effect
Busy EUR/GBP release mornings	1,848	+1.08	+2.09	−5.2	weaker, but still positive
US holidays (COMEX short session)	81	+3.14	+1.60	+2.8	possibly stronger; small sample
Other holidays and event days[a]	13–387	≥ 0[b]	–	$\lvert t\rvert < 2$	no effect

Notes:
– Basis points per day before costs, short 10:25→10:31 London.
– Raw six-minute move: bank-holiday, release-morning, US-holiday and event rows. Move with a 0.1% stop: weekday and month-day rows.
– "Others": the same trade on all remaining days. "t, diff.": t-statistic of the difference.
– On bank holidays the level itself has $t = -0.13$.
– Busy release mornings: at least five medium- or high-impact EUR or GBP releases between 07:00 and 10:31.
[a] Chinese, Japanese, Indian, German and Swiss holidays; FOMC, CPI and NFP days; GC option expiry and first notice days; 13 to 387 days each.
[b] Every group is non-negative except German Unity Day (−0.14 bp, 12 days).
Source: CME GC front contract, Databento 1-minute bars; bank holidays from gov.uk rules; event times from the MT5 economic calendar.

Three further checks:

- **Size of the overnight or pre-fix move.** Whether gold fell or rose sharply after the session open, after the London open or in the hour before the fix, the short stays positive in every bucket (correlation with the result −0.01 to −0.03). Large prior moves raise the variance, not the mean.
- **After a losing day closed by time, not by stop.** Price keeps drifting up by +0.66 bp to 10:46, but that is the general post-fix rebound seen on all days (+0.83 bp), and it is below costs. Neither a continuation long nor a re-short is justified.
- **Regime filter.** Skipping the fix when the 20-day median morning volatility is below 70 bp removes most of the quiet 2017–2019 period. In 2022 it did not help: volatility was high, but the drift was weak.

After these tests, the strategy skips English bank holidays and can optionally size Mondays up. Calendar filters on month-end or weekday are not used.

6 Implementation and live trading

The strategy runs in two forms: inside a Python multi-strategy bot on a demo account, and as a stand-alone MetaTrader 5 Expert Advisor on a live account since late September 2026. Both read the same parameters and skip the same holidays.

The CFD was chosen over the futures contract on cost. The measured GC bid-ask spread is \$0.35–0.46 per ounce, against about \$0.12–0.16 on the CFD plus \$0.06 per ounce of commission. For a trade that earns about \$0.60 per ounce on average, that difference decides whether it is profitable.

Live trades were reconciled with the backtest trade by trade, using the broker's position report and the bot's logs. Entry prices matched within the spread. The reconciliation found two engine issues, both fixed before the results above:

- Short stops were checked against the bid instead of the ask, which flattered shorts by about 0.9% of P&L.
- Missing 1-minute bars were not logged, so a skipped signal could not be explained. Every late bar now raises a warning.

The live account started in a weak week. From 28 September to 5 October 2026 the Expert Advisor made 8 trades and lost 0.69 R, or 1.9% of the account, while the backtest over the same days lost 0.61 R (Exhibit 8). Every live trade has a backtest counterpart on the same day and leg; entries differ by \$0.01–0.34.

Exhibit 8. First live week against the backtest, same days and legs

Date	Leg	Live entry	Backtest entry	Live, R	Backtest, R
28 Sep	Main short	4147.74	4147.50	+0.55	+0.49
28 Sep	Rebound long	4145.61	4145.46	+0.89	+0.90
29 Sep	Main short	4141.13	4141.07	−0.03	−0.04
30 Sep	Main short	4185.16	4185.05	−0.75	−0.75
1 Oct	Main short	4160.99	4160.93	−0.08	−0.10
2 Oct	Main short	4185.10	4185.05	−0.80	−0.78
5 Oct	Main short	4167.76	4168.10	+0.56	+0.69
5 Oct	Rebound long	4165.08	4165.09	−1.01	−1.02
Total				−0.69	−0.61

Notes:
– Live: MetaTrader 5 Expert Advisor on a live account, risk \$42–65 per trade.
– Backtest: the same engine and strategy settings as Exhibit 5, run on the same days.
– Results in R after spread and commission make the two comparable despite different position sizes.
Source: broker position report and backtest trade log.

Three data problems were found along the way, each of which would have changed the conclusions:

1. **Daylight-saving dates.** The broker server follows US dates; using European dates shifted the fix by an hour in about three weeks a year (Section 2).
2. **Futures roll inside the tick data.** In a MetaTrader 5 test on GC ticks, the continuous series switched from the expiring to the next contract within a trade on 27 roll days, creating moves of \$15–60 in six minutes. These trades added +\$33,800 of the test's +\$94,000. They are excluded from all figures here.

3. **Truncated source file.** A 30 GB exchange extract ended in April 2023 with null bytes instead of an error; it was detected by checking the last timestamp of every file.

7 Limitations and risks

The largest risk is regime: the drift was near zero for three years in a row (2017–2019) and weak for three more (2022–2024), so the strategy can go flat for long stretches without being broken.

- **Multiple testing.** More than 40 conditions were tested on the same 16 years. A few with t near 2 will be false positives; only the weaker busy-release mornings and the Monday effect clear $|t| = 3$. The holiday placebo falls just short ($t = -2.8$ for the difference from other days), but it was predicted by the mechanism, not found by searching. No part of the history was held out untouched, so the Monday refinement in particular needs confirmation on new data.
- **Short CFD sample.** The tradable backtest covers 15 months, all inside a high-volatility regime. It shows the strategy works with real costs, not what a normal year earns.
- **Unobserved mechanism.** Who trades into the fix is inferred, not seen. A change in how the auction is run or how dealers hedge could remove the effect without warning.
- **Capacity.** At a few lots the trade has no visible price impact. Its size is limited by liquidity in the six-minute window, so it suits small accounts, not a fund.
- **Execution.** A missed bar, a disconnected terminal or a server-clock error costs a whole day. The live setup checks the server offset at start-up and refuses to trade if it disagrees with the expected rule.

8 Conclusion

Gold drifts down into the LBMA AM auction, and the drift is tied to the auction itself: it is concentrated in the minutes around the 10:30 London start, partly reverses afterwards and disappears on days without an auction. It is positive in 15 of the 17 calendar years from 2010 to 2026, but its size depends heavily on the volatility regime.

On a low-cost gold CFD the effect is tradable after spread and commission. A three-leg strategy built on it returned a profit factor of 2.55 over 15 months in a strong regime, and now trades live. A realistic expectation for calm years is far lower than that period suggests.

Next steps:

- Confirm the Monday effect on data not used to find it (live months from October 2026 onward).
- Rebuild the futures tick data with one contract per day and rerun the full-history test.
- Write up the 15:00 PM auction, which shows the same drift (+1.41 bp, $t = 7.3$ on GC over 16 years) and is built but not yet traded.
- Extend the literature review to studies of the electronic LBMA auction since 2015.

References

Caminschi, A., and R. Heaney (2014). Fixing a leaky fixing: short-term market reactions to the London PM gold price fixing. *Journal of Futures Markets* 34(11), 1003–1039.

Harvey, C. R., Y. Liu, and H. Zhu (2016). ... and the cross-section of expected returns. *Review of Financial Studies* 29(1), 5–68.

SPDR Gold Trust (2026). SPDR Gold Shares (GLD): fund overview and NAV methodology. Accessed 6 October 2026.